

The CRISP-DM Process Model

Project: Real-Time Human Activity Recognition from Smartphone Sensors

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Business Understanding

1.1 Determine Business Objectives

Task Determine Business Objectives

The first objective of the analyst is to thoroughly understand, from a business perspective, what the client really wants to accomplish. Often the customer will have many competing objectives and constraints that must be properly balanced. The analyst's goal is to uncover important factors at the beginning of the project that can influence the final outcome. A likely consequence of neglecting this step would be to expend a great deal of effort producing the correct answers to the wrong questions.

Output Background

This gives details about the information that is known about the organization's business situation at the start of the project. These details will not only serve to more closely identify the business goals that will be solved, but will also serve to identify resources, both human and material, that may be used or needed during the course of the project.

This project focuses on Human Activity Recognition (HAR) using data collected from common smartphone sensors. A modern phone already includes an accelerometer and a gyroscope, so it can record motion without using an external camera, wearable device, or special lab equipment. The planned system will use the UCI HAR Dataset for training and validation, and Phyphox for collecting live sensor streams from a smartphone.

The main idea is to divide continuous sensor data into short time windows and classify each window into a human activity such as walking, sitting, standing, lying, walking upstairs, or walking downstairs. Our solution will combine classical signal processing, especially Fourier-based frequency features, with machine learning models. We will start from understandable baselines and then test a transformer-based model for sequential sensor data.

Output Business Objectives

Describe the customer's primary objective, from a business perspective, in the data mining project. In addition to the primary business objective, there are typically a large number of related business questions that the customer would like to address. For example, the primary business goal might be to keep current customers by predicting when they are prone to move to a competitor, whilst secondary business objectives might be to determine whether lower fees will affect only one particular segment of customers.

The primary business objective is to build a working prototype that can recognize a person's physical activity from smartphone sensor data in near real time. The prototype should show that useful activity monitoring can be done with existing embedded sensors, without relying on video or external tracking.

Secondary objectives are:

- Understand which sensor signals are most useful for activity classification, mainly acceleration and gyroscope signals.
- Compare simple feature-based models with a transformer-based approach.

- Create a pipeline that can later be connected to live data collected through Phyphox.
- Keep the system explainable enough for a student project, so the preprocessing, features, and evaluation are clear.

Output Business Success Criteria

Describe the criteria for a successful or useful outcome to the project from the business point of view. This might be quite specific, such as reduction of customer churn to a certain level, or general and subjective such as "give useful insights into the relationships". In the latter case it should be indicated who would make the subjective judgment.

The project will be considered successful if it produces a clear and reproducible activity classification pipeline. From the business point of view, success means that a user can collect smartphone sensor data and receive a reasonable activity prediction for each time window.

The success criteria are:

- A working data pipeline from raw or preprocessed sensor measurements to activity labels.
- A model evaluation using a train/test split with standard metrics such as accuracy, precision, recall, F1-score, and a confusion matrix.
- Evidence that the model can separate at least the main activities: walking, sitting, standing, and lying.
- A short discussion of limitations, especially differences between the UCI dataset and live Phyphox recordings.
- Code and results that are understandable and can be reproduced by another student.

1.2 Assess Situation

Task Assess Situation

This task involves more detailed fact-finding about all of the resources, constraints, assumptions, and other factors that should be considered in determining the data analysis goal and project plan.

Output Inventory of Resources

List the resources available to the project, including: Personnel (business and data experts, technical support, data mining personnel), Data (fixed extracts, access to live warehoused or operational data), Computing resources (hardware platforms), Software (data mining tools, other relevant software).

Personnel:

- Project students: responsible for data preparation, modeling, evaluation, and documentation.
- Course staff/mentor: provides guidance and feedback.

Data:

- UCI HAR Dataset, including accelerometer and gyroscope signals, engineered features, subject IDs, and activity labels.
- Optional live data collected with the Phyphox mobile application using acceleration and gyroscope sensors.

Computing resources:

- A regular laptop or desktop should be enough for baseline models.
- GPU access is helpful but not mandatory for a small transformer experiment.

Software:

- Python, pandas, NumPy, SciPy, scikit-learn, PyTorch, matplotlib/seaborn, and Jupyter Notebook.
- Phyphox app for smartphone sensor recording.

Output Requirements, Assumptions and Constraints

List all requirements of the project including schedule of completion, comprehensibility and quality of results, and security as well as legal issues. As part of this output, make sure that you are allowed to use the data.

List the assumptions made by the project. These may be assumptions about the data, which can be checked during data mining, but may also include non-checkable assumptions about the business upon which the project rests. It is particularly important to list the latter if they form conditions on the validity of the results.

List the constraints made on the project. These constraints might involve lack of resources to carry out some of the tasks in the project within the timescale required or there may be legal or ethical constraints on the use of the data or the solution needed to carry out the data mining task.

Requirements:

- Use a legal and publicly available dataset. The UCI HAR Dataset is intended for research and educational use.

- Produce a clear notebook or script that loads the data, preprocesses it, trains models, and reports results.
- Use evaluation metrics that match a multi-class classification task.
- Avoid collecting private information. Live recordings should focus only on sensor values and activity labels.

Assumptions:

- The UCI HAR labels are reliable enough for training and evaluation.
- Short time windows contain enough motion information to distinguish common activities.
- Fourier features can help capture repeated motion patterns such as walking.
- A transformer may learn useful temporal patterns if the amount of data and preprocessing are sufficient.

Constraints:

- The dataset was collected from specific subjects and one smartphone setup, so it may not generalize perfectly to all users and devices.
- The project time is limited, so the first goal is a reliable prototype rather than a production mobile app.
- Live Phyphox data may have different sampling rates, noise, and phone positions compared with the UCI dataset.

Output Risks and Contingencies

List the risks, that is, events that might occur, impacting schedule, cost or result. List the corresponding contingency plans; what action will be taken to avoid or minimize the impact or recover from the occurrence of the foreseen risks.

Main risks and contingency plans:

- Risk: The live Phyphox data format does not match the UCI feature format. Contingency: first evaluate on UCI only, then build a separate conversion step for live data.
- Risk: Transformer performance is not better than simpler models. Contingency: report it honestly and use the transformer as an experimental comparison, while keeping Random Forest/SVM/logistic regression as baselines.
- Risk: Overfitting to the UCI dataset. Contingency: use the provided train/test split, validation checks, confusion matrices, and possibly subject-based evaluation if time allows.
- Risk: Sensor noise or unstable phone placement during live collection. Contingency: collect several short recordings per activity and apply basic filtering and normalization.
- Risk: Not enough time for full real-time deployment. Contingency: demonstrate near-real-time inference on recorded windows instead of a full mobile application.

Output Terminology

Make a glossary of terminology relevant to the project. This should include at least two components:

(1) A glossary of relevant business terminology, which forms part of the business understanding available to the project.

(2) A glossary of data mining terminology, illustrated with examples relevant to the business problem in question.

Business terminology:

- Human Activity Recognition (HAR): automatic identification of a person's physical activity from sensor data.
- Smartphone sensors: built-in sensors such as accelerometer and gyroscope that measure movement and rotation.
- Real-time monitoring: producing predictions shortly after the sensor data is recorded.
- Activity label: the activity assigned to a time segment, for example walking or sitting.

Data mining terminology:

- Time window: a short segment of sensor readings used as one sample for classification.
- Feature extraction: converting raw sensor signals into useful numeric variables, such as mean, standard deviation, energy, and frequency components.
- Fourier transform: a signal-processing method that represents a time signal by its frequencies. In this project it can help describe repeated motion patterns.
- Transformer: a neural network architecture that can learn relationships across a sequence. Here it may be used on sensor windows or extracted sequences.
- Classification: predicting one label from a set of possible labels.
- Confusion matrix: a table showing which activities were predicted correctly and which activities were confused with others.

Output Costs and Benefits

Prepare a cost-benefit analysis for the project, comparing the costs of the project with the potential benefit to the business if it is successful.

Expected costs are low because the main data source is public and the required sensors already exist in a smartphone. The main costs are student time, development time, and possible computing time for transformer experiments.

Potential benefits include a working prototype for activity monitoring, better understanding of smartphone sensor data, and a reusable pipeline for future work. If the project works well, it can later support health monitoring, fitness tracking, smart-device applications, or safety-related use cases. The current project does not claim to solve these applications fully; it is a first technical step toward them.

1.3 Determine Data Mining Goals

Task Determine Data Mining Goals

A business goal states objectives in business terminology, a data mining goal states project objectives in technical terms. For example, the business goal might be "Increase catalogue sales to existing customers" while a data mining goal might be "Predict how many widgets a customer will buy, given their purchases over the past three years, relevant demographic information, and the price of the item".

Output Data Mining Goals

Describe the intended outputs of the project that will enable the achievement of the business objectives. Note that these will normally be technical outputs.

The technical goal is to build and evaluate a multi-class activity classification model using smartphone accelerometer and gyroscope data.

Planned outputs:

- A clean dataset loading and preprocessing pipeline for UCI HAR.
- A window-based representation of sensor data.
- Time-domain features such as mean, standard deviation, min, max, and signal energy.
- Frequency-domain features using Fourier transform, such as dominant frequency and spectral energy.
- Baseline classification models, for example Random Forest, SVM, logistic regression, or a simple neural network.
- A transformer-based model for sequential sensor windows, if the data format and time allow it.
- Evaluation results including accuracy, F1-score, confusion matrix, and short error analysis.
- Optional demonstration using Phyphox recordings from a smartphone.

1.4 Produce Project Plan

Task Produce Project Plan

Describe the intended plan for achieving the data mining goals and thereby achieving the business goals.

Output Project Plan

List the stages to be executed in the project, together with duration, resources required, inputs, outputs and dependencies. Wherever possible make explicit the large-scale iterations in the data mining process, for example, repetitions of the modeling and evaluation phases. As part of the project plan, it is also important to analyse dependencies between time schedule and risks. Mark results of these analyses explicitly in the project plan, ideally with actions and recommendations for actions if the risks appear.

Remember that, although this is the only task in which the Project Plan is directly named, nevertheless it should be continually consulted and reviewed throughout the project. This should be a document which should at least be consulted whenever a new Task is started or a further iteration of a task or activity is begun.

Proposed project plan:

- Stage 1 - Problem definition and data review: understand the UCI HAR files, activity labels, sensor channels, and project requirements. Output: short data summary.
- Stage 2 - Data preparation: load train/test files, check shapes, labels, missing values, and normalize data if needed. Output: clean training and test datasets.
- Stage 3 - Feature engineering: compute or organize time-domain features and Fourier-based frequency features. Output: feature table ready for modeling.
- Stage 4 - Baseline modeling: train simple models and compare results. Output: baseline performance report.
- Stage 5 - Transformer experiment: prepare sequential windows and train a small transformer model. Output: comparison with baseline models.
- Stage 6 - Live-data compatibility check: collect a small Phyphox sample and test the preprocessing pipeline. Output: notes about what works and what needs adjustment.
- Stage 7 - Final evaluation and documentation: prepare final metrics, plots, limitations, and conclusions. Output: final report and code.

Dependencies: modeling depends on successful data loading and preprocessing. The live-data demonstration depends on matching Phyphox sensor streams to the training format. If time becomes limited, the transformer and live-data parts will be reduced, while the UCI evaluation will remain the core deliverable.

Output Initial Assessment of Tools and Techniques

At the end of the first phase, the project also performs an initial assessment of tools and techniques. Here, you select a data mining tool which supports various methods for different stages of the process, for example. It is important to assess tools and techniques early in the process since the selection of tools and techniques possibly influences the entire project.

Initial tools and techniques:

- Python will be used as the main development language.
- pandas and NumPy will be used for data loading and manipulation.
- SciPy will be used for signal processing and Fourier transform features.
- scikit-learn will be used for baseline models and evaluation metrics.
- PyTorch will be used for the transformer model if time allows.
- matplotlib or seaborn will be used for plots, including confusion matrices and signal examples.
- Phyphox will be used to record live accelerometer and gyroscope data from a smartphone.

The first modeling approach should be simple and reliable. After that, the project can test whether Fourier-based features and a transformer model add value over the baseline.